What if bins could guide collection decisions?

From sensor readings to smarter collection thresholds



DATA → PROBABILITY → SIMULATION → POLICY INSIGHT

INDIVIDUAL PROJECT

The trade-off every smart city faces

COLLECT EARLY

More collection activity • Less high-fullness exposure

COLLECT LATE

Fewer collections • More high-fullness exposure

Question: where should the collection trigger sit?

INDIVIDUAL PROJECT

One dataset. A reproducible evidence pipeline.

01 AUDIT → **02** CLEAN → **03** TRANSITIONS → **04** RVs → **05** SIMULATE

Historical smart-bin observations → daily state behaviour → simulated policies

INDIVIDUAL PROJECT

The data audit changes what we can trust

766

Exact duplicate records removed

30,661

Master records after deduplication

32

Bins used for behaviour modelling

Only consecutive-day observations inform state transitions.

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Two uncertainties power each simulated day

RV 1

Does fullness increase?

Probability depends on current fullness state

RV 2

If yes, by how much?

Increase size is sampled from observed changes

INDIVIDUAL PROJECT

Three collection triggers. Same simulation rules.



6

Earlier trigger



8

Middle trigger

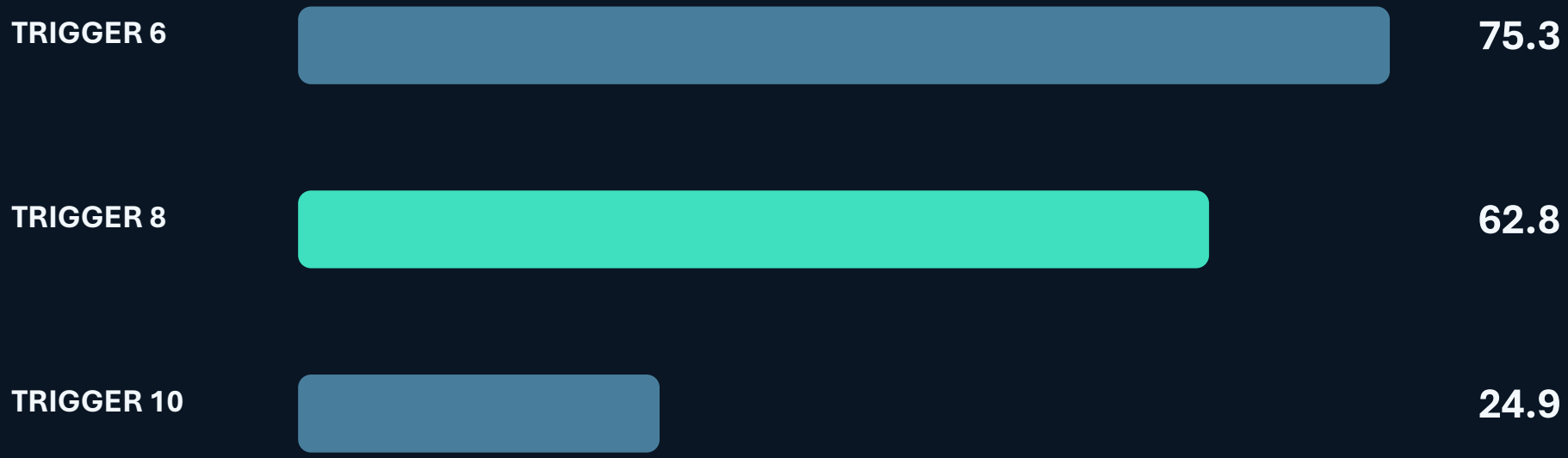


10

Later trigger

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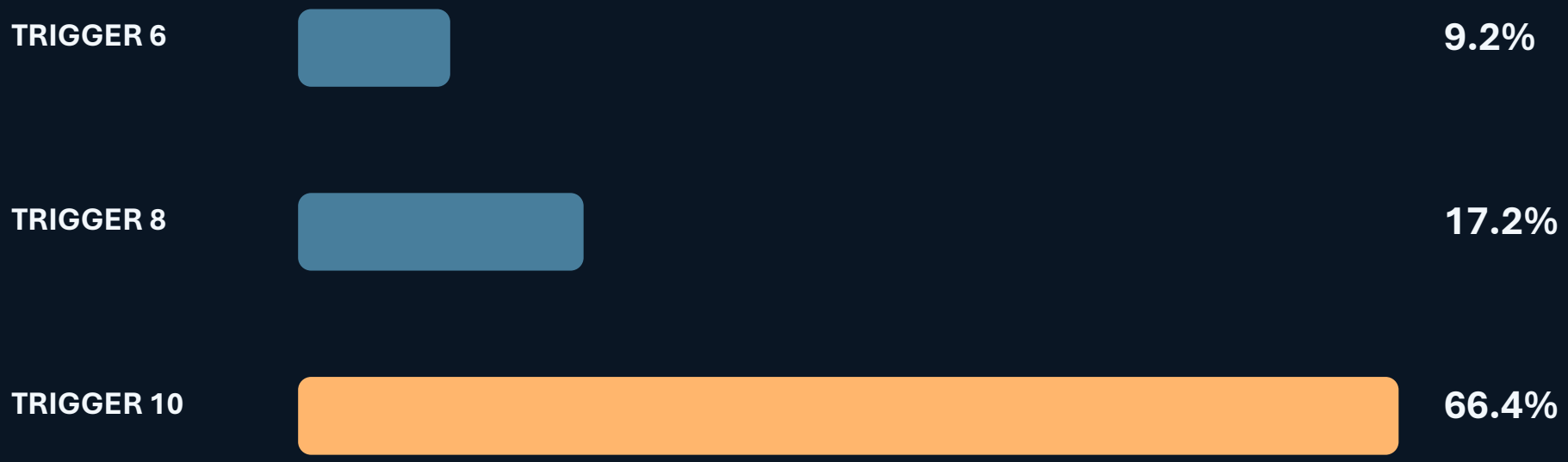
Fewer collections? The difference is dramatic.



Average simulated collections per 365-day year

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But fewer collections mean fuller bins for longer



Share of simulated days at high fullness (states 8 or 10)

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The model is a decision aid—not a field verdict

SUPPORTED

Thresholds trade collection frequency against high-fullness exposure.

Sensitivity tests compare lower, base and higher accumulation.

NOT YET VALIDATED

Real collection events, service delays, route costs and overflow outcomes require external validation.

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From simulation to a real smart-city pilot

1

VERIFY SENSOR DATA

2

VALIDATE COLLECTION LOGS

3

ADD COST + SERVICE DELAYS

4

PILOT & MONITOR

The goal is a better-informed trigger—not a claim of proven operational savings.